

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

In recent years, continuous health monitoring has become increasingly important due to the rising prevalence of stress-related disorders, fatigue, and chronic medical conditions. Early detection of physiological changes such as abnormal heart rate, reduced blood oxygen levels, and fluctuations in body temperature can significantly improve preventive healthcare outcomes. However, conventional monitoring systems are often limited to hospital environments, are bulky in design, or require frequent manual observation, making them unsuitable for long-term and continuous use. As a result, there is a growing demand for compact, portable, and cost-effective systems that can continuously log vital health parameters in real time.

Advancements in embedded systems and Internet of Things (IoT) technologies have enabled the development of smart wearable devices capable of real-time physiological monitoring. Microcontrollers with integrated wireless communication features allow seamless data acquisition, processing, storage, and remote transmission. By combining compact sensors with low-power processing units, it is possible to design systems that provide reliable and continuous health tracking without causing discomfort to the user. Such systems play a crucial role in early fatigue detection, stress monitoring, elderly care, and remote patient supervision.

The proposed Compact Health Parameter Recorder addresses these challenges by integrating a low-power microcontroller with advanced biomedical sensors for heart rate, blood oxygen saturation (SpO_2), and body temperature measurement. The system continuously collects physiological signals, processes them to ensure accuracy, and stores or transmits the data for long-term analysis. Continuous logging enables trend-based evaluation, allowing the identification of gradual deviations from normal health patterns that may indicate fatigue or the onset of medical conditions. This approach shifts healthcare from reactive treatment to proactive and preventive monitoring.

Furthermore, the compact design and wireless capability make the system suitable for wearable applications in daily life. It can be used for monitoring athletes, industrial workers, elderly individuals, and patients requiring long-term supervision. The low power consumption ensures extended operation, enhancing practicality for continuous use. By combining affordability, portability, and reliable performance, the proposed system contributes to accessible preventive healthcare and supports the growing need for smart, connected health monitoring solutions.

1.2 BACKGROUND

The growing incidence of lifestyle-related disorders, occupational fatigue, and chronic diseases has increased the need for continuous health monitoring systems. Traditional health monitoring methods are primarily hospital-based and depend on periodic clinical measurements, which may not capture gradual physiological changes occurring over time. Parameters such as heart rate, blood oxygen saturation (SpO_2), and body temperature are critical indicators of a person's overall health condition. Variations in these parameters can signal fatigue, stress, respiratory issues, infections, or cardiovascular abnormalities. However, without continuous logging, early warning signs are often missed.

Recent advancements in wearable technology and embedded systems have made it possible to monitor vital signs outside clinical settings. Low-power microcontrollers and compact biomedical sensors now enable real-time acquisition and processing of physiological data in portable devices. Wireless communication technologies further allow remote data transmission for long-term storage and analysis. Despite these developments, many existing systems remain expensive, bulky, or limited in functionality. Therefore, there is a need for an integrated, compact, and cost-effective solution that supports continuous monitoring and long-term trend analysis for preventive healthcare applications.

CHAPTER 2

LITERATURE SURVEY

Sadeghi. H et al. (2025) developed a multi-vital wearable system capable of monitoring heart rate (HR), SpO₂, body temperature, and respiratory rate, integrated with embedded artificial intelligence for real-time anomaly detection. The study implemented lightweight machine learning models directly on the device to perform edge inference, reducing reliance on cloud processing and minimizing data transmission requirements. The authors emphasized sensor fusion techniques to improve detection accuracy by combining multiple physiological parameters. Their work demonstrates how multimodal data integration and low-complexity on-device algorithms can enhance reliability and responsiveness, making it highly relevant for incorporating fatigue classification features into compact ESP32-based health monitoring systems.

Zeng. C et al. (2024) investigated driver fatigue detection using short 2-minute ECG segments and extracted heart rate variability (HRV) features to classify alert and fatigued states. The study demonstrated that time-domain and frequency-domain HRV parameters can effectively discriminate fatigue conditions with good classification accuracy. The authors also analyzed sex-based physiological differences, emphasizing the importance of personalized fatigue detection models. This work highlights that compact, short-window HRV analysis can be implemented in embedded systems for real-time fatigue monitoring.

Srinivasan. A. G et al. (2024) investigated the relationship between multiple heart rate variability (HRV) indices and multidimensional fatigue models, analyzing time-domain, frequency-domain, and nonlinear HRV parameters. The study demonstrated that specific HRV features, such as reduced RMSSD and altered LF/HF ratios, are strongly associated with increased fatigue levels. The authors emphasized integrating physiological metrics with subjective assessments and performance indicators to improve detection accuracy. Their findings suggest that relying on a

single parameter may be insufficient, and combining HR/HRV trends with additional vital signals enhances robustness. This work supports the integration of HRV with SpO₂ and temperature monitoring in wearable systems for more reliable and comprehensive fatigue detection.

Contoli. C (2024) presented a comprehensive survey on energy-aware wearable activity recognition systems, emphasizing techniques to optimize power consumption without significantly compromising data accuracy. The study discussed strategies such as duty-cycling sensors, adaptive sampling based on activity levels, and selective feature extraction to reduce computational load. It also highlighted the importance of minimizing wireless transmission frequency by batching data and performing local preprocessing. These approaches are particularly relevant for ESP32-based wearable health recorders, where efficient power management is essential for enabling long-term continuous monitoring with extended battery life.

Tesema. W (2024) presented a detailed taxonomy of low-power design techniques for wearable devices, focusing on strategies to extend operational lifetime in continuous monitoring applications. The study reviewed methods such as batching wireless transmissions to reduce radio usage, implementing sensor duty cycling to minimize active time, and performing local preprocessing to decrease data volume before transmission. It also discussed dynamic power management and sleep-mode utilization in embedded systems. These design patterns are directly applicable to long-duration SpO₂ and heart rate logging systems using microcontrollers like the ESP32, where efficient energy management is critical for sustained wearable health monitoring.

Rehman. R. Z. U et al. (2024) conducted a systematic comparison between heart rate (HR) and heart rate variability (HRV) metrics derived from photoplethysmography (PPG) signals and those obtained from electrocardiogram (ECG) recordings under free-living conditions. The study evaluated time-domain and frequency-domain HRV parameters and analyzed the impact of motion artifacts and

signal noise on measurement accuracy. The authors found that while PPG-based HR estimation is generally reliable, certain HRV indices may diverge from ECG-derived values, especially during high-motion activities. The paper emphasizes the need for careful signal preprocessing and validation when using PPG sensors like the MAX30102 for HRV-based fatigue analysis in wearable systems.

Kang. M (2024) reviewed recent advancements in energy harvesting technologies and power management modules for wearable electronics, focusing on methods to extend device operating time. The study surveyed various energy sources such as thermoelectric, piezoelectric, and solar harvesters, along with efficient power management integrated circuits (PMICs) designed for low-power systems. It emphasized hybrid energy harvesting combined with aggressive power optimization strategies, including dynamic voltage scaling and sleep-mode utilization. The paper highlighted how integrating efficient energy harvesting with intelligent power control can significantly enhance continuous monitoring runtime, making it particularly relevant for ultra-long-duration wearable health monitoring systems.

Kim. K. B (2023) presented a comprehensive review of photoplethysmography (PPG) technology for wearable applications, discussing the fundamental operating principles, hardware configurations, and signal-processing challenges. The paper details motion artifact reduction techniques, filtering methods, and calibration strategies essential for improving SpO₂ and heart rate accuracy. The review provides practical insights directly applicable to MAX30102-based monitoring systems, especially in wearable and low-power environments.

Nardelli. M (2023) reviewed recent advancements in wearable photoplethysmography (PPG) systems, highlighting improvements in sensor miniaturization, low-power optical designs, and enhanced analog front-end circuits. The study discussed practical deployment challenges such as sensor placement variability, motion artifacts during daily activities, and the need for robust algorithms that can operate under noisy conditions. It emphasized adaptive filtering, peak

detection optimization, and lightweight feature extraction techniques suitable for real-time processing. Importantly, the review outlined computationally efficient methods that can be implemented on resource-constrained microcontrollers like the ESP32, making it highly relevant for embedded wearable health monitoring applications.

Scardulla. F (2023) examined the potential and limitations of wearable photoplethysmography (PPG) sensors, focusing on their growing use in consumer and clinical health monitoring devices. The study identified key challenges such as motion artifacts, variability in sensor placement, differences in skin tone, and inconsistencies in signal processing methods across studies. A major concern highlighted was the lack of standardized data collection procedures and validation frameworks, which can lead to inconsistent performance evaluation. The author recommended structured validation protocols, controlled testing conditions, and benchmark datasets to improve reliability and comparability. This guidance is valuable for designing systematic experiments and establishing long-term logging standards in compact health monitoring systems.

Chen. Y. X (2023) proposed a fatigue detection method based on peak-feature analysis of photoplethysmography (PPG) signals, extracting parameters such as peak intervals, amplitude variations, and dicrotic notch characteristics. The study demonstrated a significant correlation between these extracted features and subjective fatigue assessment questionnaires, validating the physiological relevance of the proposed index. The author emphasized that computationally lightweight feature extraction techniques can be implemented in real time without requiring complex machine learning models. This approach makes it suitable for embedded platforms like ESP32, enabling efficient on-device fatigue scoring using MAX30102-based PPG signals in wearable monitoring systems.

Lee. J et al. (2023) proposed a multimodal physiological monitoring framework for fatigue and stress detection by combining heart rate variability (HRV), blood oxygen saturation (SpO_2), skin temperature, and respiratory rate signals. The study implemented feature-level fusion techniques and applied lightweight machine learning

classifiers to improve detection accuracy compared to single-parameter approaches. Experimental results showed that integrating multiple vital signals significantly enhanced sensitivity and specificity in identifying fatigue states. The authors emphasized that multimodal fusion reduces false positives caused by noise or temporary fluctuations in individual signals. This work strongly supports the integration of MAX30102-derived PPG features with DS18B20 temperature data in embedded wearable systems, enabling more reliable and robust long-term fatigue monitoring using on-device processing.

Huhn. S et al. (2022) reviewed the growing role of affordable wearable technologies in health research and large-scale population studies. The paper discussed validation strategies to ensure measurement accuracy, emphasizing comparison with clinical-grade reference devices and standardized testing protocols. It also addressed important considerations such as data privacy, ethical data handling, and secure wireless transmission in wearable health systems. The authors highlighted challenges in real-world deployment, including user compliance, long-term reliability, and environmental influences on sensor performance. Their recommendations provide essential methodological guidance for conducting field trials and validating compact health monitoring recorders in practical healthcare settings.

Contardi. U. A (2021) developed a continuous heart rate and SpO₂ monitoring system by integrating the MAX30102 sensor with an ESP32 microcontroller and a webserver interface. The study explained I²C-based sensor interfacing, real-time data acquisition, and wireless transmission for remote monitoring. Low-power optimization techniques were applied to extend operating time, making the system suitable for wearable applications. Their implementation provides a practical framework for firmware design, IoT connectivity, and long-term vital sign logging in compact health monitoring devices.

Castañeda. D et al. (2018) provided a comprehensive review of wearable photoplethysmography (PPG) sensors, discussing advancements in optical sensor design, signal acquisition methods, and integration into portable health devices. The study highlighted major real-world challenges such as motion artifacts, poor skin contact, ambient light interference, and calibration errors that can significantly affect measurement accuracy. It emphasized the need for proper sensor placement, mechanical stability, and effective digital filtering techniques to improve signal reliability. The review also compared reflective and transmissive PPG configurations, offering insights into practical wearable implementations. This work provides essential background for understanding performance limitations and the importance of careful hardware and signal-processing design when using sensors like the MAX30102 in continuous health monitoring system.

CHAPTER 3

EXISTING & PROPOSED METHOD

3.1 EXISTING METHOD

Existing health monitoring methods primarily rely on hospital-based equipment and periodic clinical measurements to assess vital parameters such as heart rate, oxygen saturation (SpO₂), and body temperature. Traditional systems use standalone medical devices like pulse oximeters, ECG monitors, and digital thermometers, which require manual operation and do not support continuous long-term logging. These systems are generally bulky, expensive, and limited to short-duration measurements under clinical supervision.

In recent years, wearable devices such as smartwatches and fitness bands have introduced continuous heart rate monitoring using photoplethysmography (PPG). However, many of these consumer-grade devices focus mainly on activity tracking and basic health metrics, offering limited access to raw physiological data for detailed fatigue or stress analysis. Additionally, most existing wearable systems depend heavily on cloud processing, increasing power consumption and reducing battery life during continuous monitoring.

Some IoT-based health monitoring systems have been developed using microcontrollers and biomedical sensors to transmit real-time data wirelessly. While these systems improve portability, they often lack integrated multi-parameter analysis and structured long-term data logging for fatigue detection. Many designs also face challenges such as motion artifacts, power inefficiency, limited onboard processing, and lack of standardized validation procedures.

Overall, the existing methods provide basic vital monitoring but do not fully address the need for a compact, low-power, integrated system capable of continuous long-term health logging and fatigue analysis in a wearable form factor.

3.2 PROPOSED METHOD

The proposed method introduces a Compact Health Parameter Recorder designed for continuous and long-term monitoring of vital signs with integrated fatigue analysis. The system is built around an ESP32 microcontroller, which serves as the central processing and communication unit. Physiological signals are acquired using the MAX30102 sensor for heart rate and SpO₂ measurement and the DS18B20 digital sensor for body temperature monitoring. These sensors are interfaced with the microcontroller through efficient communication protocols to ensure reliable real-time data acquisition.

The collected signals undergo preprocessing within the ESP32 to remove noise and motion artifacts using digital filtering techniques. Heart rate is calculated in beats per minute (BPM), oxygen saturation is computed as a percentage value, and temperature readings are recorded in degrees Celsius. In addition, trend-based analysis is performed to monitor variations over time. Fatigue detection is implemented by analyzing deviations in resting heart rate, HRV-related patterns (if derived), SpO₂ fluctuations, and abnormal temperature changes. A threshold-based or lightweight rule-based algorithm is used for on-device decision making to minimize computational complexity and power consumption.

To support long-term monitoring, the processed data is stored locally and can be transmitted wirelessly via Wi-Fi or Bluetooth for remote access and cloud-based analysis. Power optimization techniques such as sensor duty cycling, adaptive sampling, and microcontroller sleep modes are incorporated to extend battery life. The compact and wearable design ensures user comfort during continuous use. Overall, the proposed method provides an integrated, low-power, and cost-effective solution for real-time vital monitoring and early fatigue detection in wearable healthcare applications.

3.2.1 BLOCK DIAGRAM

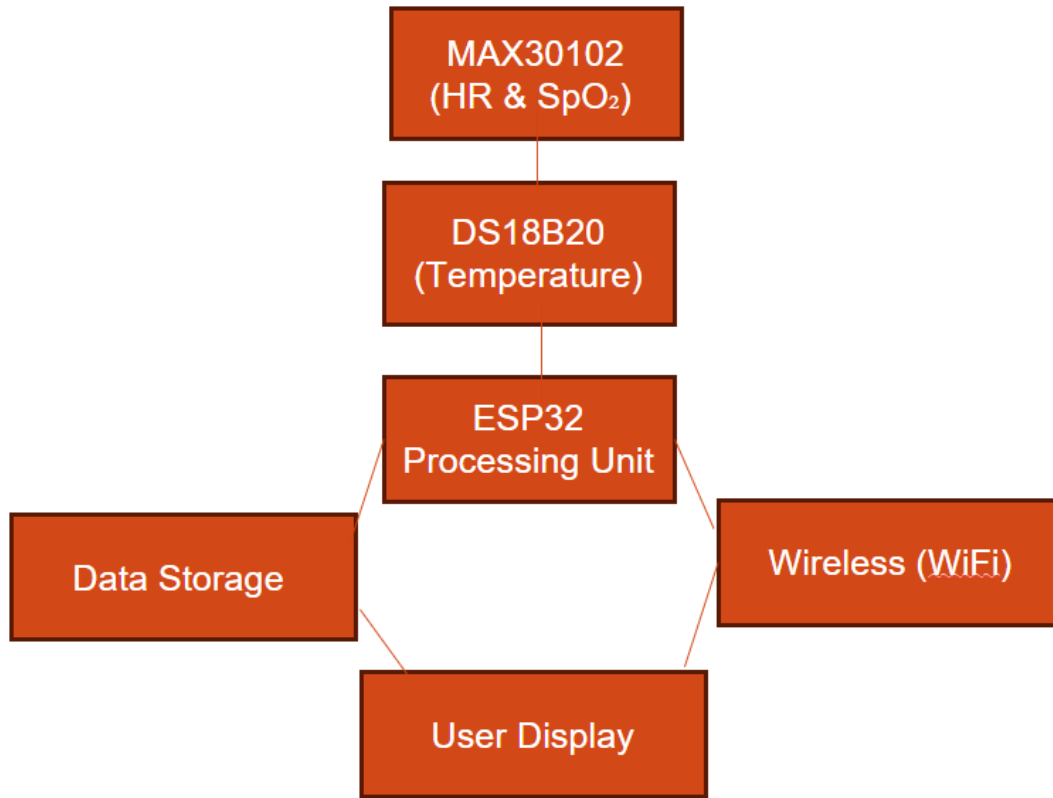


Figure 3.1 Block Diagram of Proposed Method

MAX30102 (HR & SpO₂):

The MAX30102 sensor is used to measure heart rate and blood oxygen saturation (SpO₂) using optical photoplethysmography (PPG) technology. It emits red and infrared light into the skin and detects variations in reflected light caused by blood flow changes. The sensor sends digital data to the ESP32 via I²C communication, allowing accurate calculation of heart rate in BPM and oxygen levels in percentage. It plays a key role in fatigue and health monitoring.

DS18B20 (Temperature):

The DS18B20 is a digital temperature sensor used to measure body temperature accurately. It communicates with the ESP32 using a one-wire protocol and provides

calibrated digital output without requiring additional analog-to-digital conversion. The temperature readings help identify abnormal thermal variations related to fever, stress, or fatigue.

ESP32 Processing Unit:

The ESP32 acts as the central controller of the system. It collects data from both sensors, performs signal filtering to reduce noise, calculates heart rate and SpO₂ values, and processes temperature readings. It also executes fatigue detection logic, manages data storage, and controls wireless communication. Its built-in Wi-Fi capability enables IoT-based health monitoring.

Data Storage:

The data storage unit stores the processed physiological parameters for long-term analysis. The ESP32 can save time-stamped data in internal memory or external storage such as an SD card. This enables trend monitoring over extended periods, which is essential for detecting fatigue patterns and gradual health changes.

Wireless (WiFi):

The wireless module, integrated within the ESP32, enables data transmission to a cloud server, web dashboard, or mobile application. This allows remote monitoring and real-time health tracking. Wireless connectivity supports IoT functionality and enhances accessibility for caregivers or healthcare professionals.

User Display:

The user display, such as an OLED or LCD screen, shows real-time values of heart rate, SpO₂, and body temperature. It provides immediate feedback to the user and can also display warning messages if abnormal values are detected. This ensures convenient and instant access to health information without needing external devices.

3.2.2 CIRCUIT DIAGRAM

The circuit diagram represents the hardware implementation of the Compact Health Parameter Recorder, designed to monitor heart rate, SpO₂, and body temperature using an ESP32-based embedded system. The ESP32-WROOM-32 module acts as the central processing unit and is responsible for data acquisition, signal processing, storage control, and wireless communication.

The MAX30102 sensor is interfaced with the ESP32 using the I²C communication protocol through the SDA and SCL lines. It operates at 3.3V and internally contains red and infrared LEDs along with a photodetector to measure blood volume variations using photoplethysmography (PPG). The sensor captures optical signals reflected from the skin and converts them into digital data, which is transmitted to the ESP32. The ESP32 processes this data to calculate heart rate in beats per minute (BPM) and blood oxygen saturation (SpO₂) percentage. Proper power connections and grounding are maintained to reduce signal noise and improve measurement accuracy.

The DS18B20 temperature sensor is connected to the ESP32 using a one-wire communication interface through the DQ pin. A 4.7 k Ω pull-up resistor is connected between the data line and VDD to ensure stable communication. The sensor provides calibrated digital temperature readings, eliminating the need for additional analog signal conditioning. This simplifies the circuit and improves reliability.

A voltage regulator section provides a stable 3.3V supply to the ESP32 and sensors. Maintaining a regulated voltage is essential for accurate sensor performance and stable wireless communication. Decoupling capacitors (not explicitly shown but typically included) help suppress voltage fluctuations and electrical noise.

A push-button switch is included for reset or user control functions, allowing manual restarting or mode selection. The header connector enables programming, debugging, or external interfacing. The power input connector supports battery or external DC supply, making the system portable and suitable for wearable applications.

Overall, the circuit integrates sensing, processing, and communication modules into a compact and efficient design. The use of digital communication protocols (I²C and one-wire) reduces wiring complexity and enhances reliability. The design ensures low power consumption, accurate signal acquisition, and seamless wireless transmission, making it suitable for continuous long-term health monitoring.

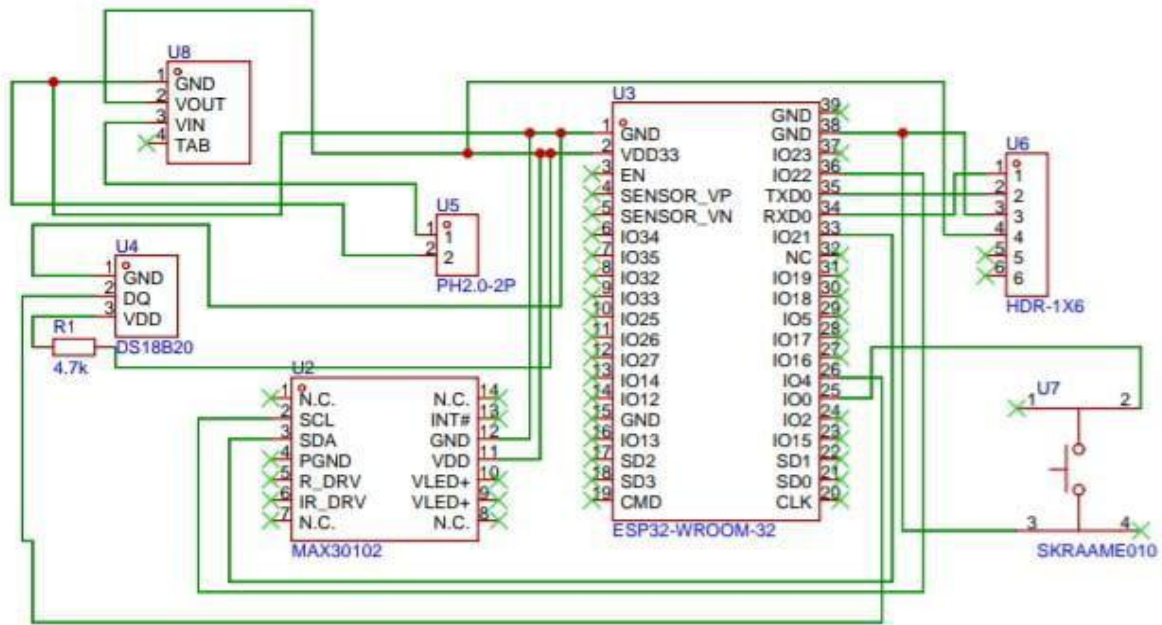


Figure 3.2 Circuit Diagram of Proposed Method

3.2.3 WORKING PRINCIPLE

The Compact Health Parameter Recorder functions as a continuous monitoring system that measures vital parameters such as heart rate, blood oxygen saturation (SpO₂), and body temperature. When power is supplied, the ESP32 microcontroller initializes the system and establishes communication with all connected sensors. The MAX30102 sensor measures heart rate and SpO₂ by detecting variations in blood flow using optical signals, while the DS18B20 sensor measures body temperature through a digital one-wire interface. The system begins data acquisition immediately and ensures synchronized communication between all modules for smooth operation.

The ESP32 collects this data in real time and processes it by applying filtering techniques to remove noise and improve accuracy. It then calculates heart rate in beats per minute (BPM), SpO₂ percentage, and temperature values. Based on these readings, the system performs simple fatigue analysis by identifying abnormal trends such as increased heart rate, decreased oxygen levels, or unusual temperature changes. Threshold-based logic is used to detect critical conditions, and alerts can be generated when values exceed safe limits.

The processed data is displayed on a user interface like an OLED or LCD screen for immediate viewing. Simultaneously, the data is stored in an SD card with time stamps, enabling long-term health tracking and analysis. In addition to local storage, the ESP32 uses its built-in WiFi to transmit the data to a cloud server or mobile application, allowing remote monitoring, visualization through graphs, and alert notifications for abnormal conditions. This ensures accessibility of health data from anywhere.

To ensure efficient operation, power-saving techniques such as sensor duty cycling and ESP32 sleep modes are implemented, making the system suitable for long-duration wearable use. The compact design and low power consumption enhance portability and user comfort. Overall, the working model demonstrates an integrated approach combining

sensing, processing, storage, and communication for reliable, efficient, and continuous health monitoring, supporting early detection of fatigue and preventive healthcare management.

3.2.4 ADVANTAGES

- Compact and portable design suitable for wearable applications
- Provides continuous real-time monitoring of vital parameters
- Low power consumption for long-duration operation
- Supports wireless data transmission for remote monitoring
- Enables long-term data storage and trend analysis
- Cost-effective compared to traditional medical equipment
- Easy integration with IoT platforms and cloud systems
- Early detection of fatigue and health abnormalities
- User-friendly interface for easy understanding of data
- Scalable system that can be enhanced with additional sensors

3.2.5 APPLICATIONS

- Remote patient monitoring allows continuous tracking of vital signs like heart rate, SpO₂, and temperature from home, enabling real-time supervision by doctors and reducing hospital visits.
- Elderly monitoring provides constant health tracking for senior citizens, detecting abnormalities early and alerting caregivers for quick response and improved safety.
- Athlete monitoring helps track performance, fatigue, and recovery, ensuring optimal training and preventing overexertion or injuries.
- Industrial worker safety monitoring detects fatigue, stress, and low oxygen levels in hazardous environments, improving safety and reducing accidents.
- Preventive healthcare enables early detection of health issues by analyzing long-term data trends, supporting timely medical intervention.

- Home healthcare systems provide a low-cost and convenient solution for continuous monitoring without the need for hospital equipment.
- Fitness and wellness tracking helps individuals monitor daily health parameters and maintain a healthy lifestyle.
- Telemedicine support allows doctors to remotely access patient data and provide consultation without physical presence.
- Rehabilitation monitoring helps track patient recovery progress after surgery or illness by analyzing vital signs over time.
- Smart wearable devices integration enables the system to be used in compact wearable forms like wristbands for daily health monitoring.

CHAPTER 4

HARDWARE & SOFTWARE REQUIREMENTS

4.1 HARDWARE REQUIREMENTS

4.1.1 MICROCONTROLLER

The microcontroller used in this project is the ESP32, which acts as the central processing unit of the system. It is responsible for collecting data from sensors such as the MAX30102 and DS18B20, processing the signals, and calculating parameters like heart rate, SpO₂, and body temperature. The ESP32 also controls peripheral components such as the display, data storage unit, and wireless communication, ensuring smooth coordination between all modules. Its real-time processing capability allows continuous monitoring and analysis of physiological data below in Figure 4.1.



Figure 4.1 Microcontroller

The ESP32 is selected because it includes built-in WiFi and Bluetooth, enabling wireless data transmission without the need for external modules. It operates at 3.3V, which is compatible with most sensors used in the system, and supports multiple communication protocols such as I²C, SPI, and UART for easy interfacing. Additionally, it offers low power consumption with sleep modes, making it suitable for wearable and long-term monitoring applications. Its compact size, cost-effectiveness, and strong community support further make it an ideal choice for IoT-based health monitoring systems.

4.1.2 MAX30102 SENSOR MODULE

The MAX30102 is an advanced optical sensor used for measuring heart rate and blood oxygen saturation (SpO₂) using the photoplethysmography (PPG) technique. It consists of integrated red and infrared LEDs, a photodetector, optical elements, and low-noise signal processing circuitry. The sensor emits light into the skin and measures the variations in reflected light caused by changes in blood volume during each heartbeat. These variations are converted into digital signals and transmitted to the ESP32 via the I²C communication interface, where further processing is performed to calculate heart rate and oxygen saturation levels.

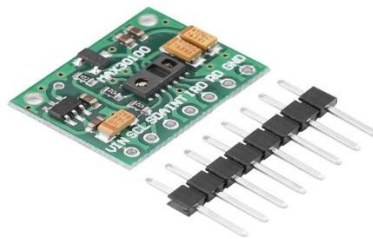


Figure 4.2 MAX30102 Sensor Module

The sensor is highly suitable for wearable and continuous health monitoring applications due to its compact size, low power consumption, and high sensitivity. It includes features such as ambient light cancellation, programmable LED current, and adjustable sampling rates, which enhance measurement accuracy under different conditions. Proper placement and stable contact with the skin are important to reduce motion artifacts and improve signal quality. The MAX30102 enables reliable real-time monitoring of vital parameters, making it a key component in detecting fatigue and supporting preventive healthcare systems.

4.1.3 DS18B20 TEMPERATURE SENSOR

The DS18B20 is a digital temperature sensor used to measure body temperature accurately in embedded systems. It operates using a one-wire communication protocol, which means it requires only a single data line along with power and ground connections. The sensor provides calibrated digital output, eliminating the need for external analog-to-digital conversion. It can measure temperature over a wide range with an accuracy of about $\pm 0.5^{\circ}\text{C}$, making it suitable for health monitoring applications.



Figure 4.3 DS18B20 Temperature Sensor

This sensor is compact, low-power, and easy to interface with microcontrollers like the ESP32. It supports multiple sensors on the same data line, which adds flexibility in system design. A pull-up resistor (typically $4.7\text{ k}\Omega$) is required to ensure stable communication. The DS18B20 is reliable for continuous temperature monitoring and helps in detecting abnormal changes such as fever or stress-related variations, making it an important component in the overall health monitoring system.

4.1.4 MICRO SD CARD MODULE

The Micro SD Card Module is used to interface a micro SD card with the ESP32 for storing vital health data such as heart rate, SpO₂, and temperature readings. It communicates with the microcontroller using the SPI protocol, ensuring fast and reliable data transfer. The module includes essential circuitry like voltage regulation and, in some cases, level shifting, making it compatible with 3.3V systems like the ESP32. It connects using SPI pins such as MISO, MOSI, SCK, and CS, allowing simple integration into the circuit.

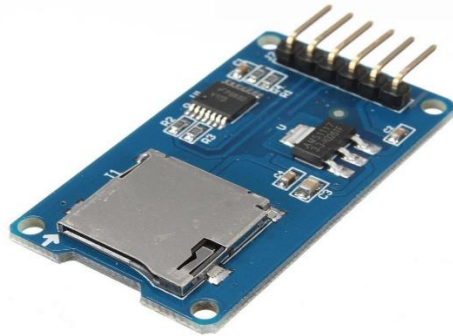


Figure 4.4 Micro SD Card Module

This module enables efficient long-term data logging by storing time-stamped records of physiological parameters in a structured format. The stored data can be accessed later for detailed analysis, trend monitoring, and identifying fatigue or abnormal health conditions. It supports various SD card capacities, allowing flexibility based on storage needs. Additionally, it reduces dependency on continuous wireless transmission by providing local backup of data. Its compact design, low power usage, and reliability make it an essential component for continuous and wearable health monitoring systems.

4.1.5 MICRO SD CARD

The Micro SD Card is used as the primary storage medium for recording vital health data such as heart rate, SpO₂, and body temperature. It works with the Micro SD Card Module to store data received from the ESP32 in a structured format with time stamps. This enables continuous and organized logging of physiological parameters over long durations, which is important for analyzing health trends and fatigue patterns.



Figure 4.5 Micro SD Card

The card provides large storage capacity (such as 4GB or higher), making it suitable for storing extensive health records without frequent data clearing. The stored data can be easily accessed, transferred, and analyzed using a computer or mobile device. It is compact, lightweight, and consumes very low power, which makes it ideal for portable and wearable systems. Additionally, it ensures data reliability by acting as a local backup, so information is not lost even if there is no internet or WiFi connection during monitoring.

4.1.6 3.7V LI-PO BATTERY (600–800MAH)

The 3.7V Lithium-Polymer (Li-Po) battery is used as the primary power source for the Compact Health Parameter Recorder, supplying stable and rechargeable power to the ESP32, sensors, and other connected components. It is widely preferred in portable and wearable devices due to its high energy density, lightweight structure, and compact size. Unlike conventional batteries, Li-Po batteries can be easily integrated into small form-factor designs, making them ideal for continuous health monitoring applications. With a capacity of 600–800mAh, the battery is capable of supporting long-duration operation depending on system load and usage conditions. It can be recharged using a TP4056 charging module, which provides protection against overcharging, over-discharging, and short circuits, ensuring safe operation. The battery works efficiently with the ESP32's power-saving features such as deep sleep and light sleep modes, helping to significantly extend battery life during continuous monitoring.



Figure 4.6 3.7V Li-Po Battery (600–800mAh)

Additionally, the battery provides stable voltage output, which is essential for accurate sensor readings and reliable system performance. It supports multiple charge cycles, making it cost-effective and reusable for long-term use. Its portability allows the system to be used in various real-world scenarios such as wearable devices, remote monitoring, and outdoor applications. Overall, the Li-Po battery ensures uninterrupted operation, mobility, and efficiency of the health monitoring system.

4.1.7 TP4056 CHARGING MODULE

The TP4056 charging module is used to safely charge the 3.7V Li-Po battery in the system. It is a linear charging controller designed specifically for single-cell lithium-ion or lithium-polymer batteries. The module takes input power (typically from a USB source) and regulates the charging process to provide a constant current and constant voltage (CC/CV) charging method, ensuring safe and efficient battery charging.



Figure 4.7 TP4056 Charging Module

The TP4056 module includes built-in protection features such as overcharge protection, over-discharge protection, and short-circuit protection, which help prevent battery damage and improve safety. It also has indicator LEDs to show charging status, such as charging and fully charged conditions. The module is compact, low-cost, and easy to integrate into the circuit, making it ideal for portable and wearable applications. By using this module, the system can be conveniently recharged and used repeatedly, ensuring reliable long-term operation.

4.1.8 4.7K Ω RESISTORS

The 4.7k Ω resistors are used as pull-up resistors in the circuit to ensure stable and reliable communication between the ESP32 and the connected sensors. They are connected to the I²C communication lines (SDA and SCL) of the MAX30102 sensor and the one-wire data line (DQ) of the DS18B20 temperature sensor. These resistors keep the signal lines at a defined high logic level when no device is actively transmitting, preventing floating or undefined states in digital communication. They play a crucial role in maintaining signal integrity by ensuring proper voltage levels and reducing the chances of communication errors. In digital protocols like I²C and one-wire, pull-up resistors are essential for correct data transmission and synchronization between devices. The 4.7k Ω value is widely used because it offers a good balance between current consumption and signal rise time, ensuring efficient operation without excessive power loss.

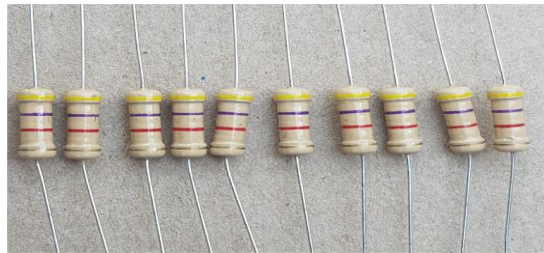


Figure 4.8 4.7K Ω resistors

Additionally, these resistors help improve system stability, especially in environments where electrical noise or long wiring may affect signals. They ensure smooth data exchange between the microcontroller and sensors, leading to accurate measurement of vital parameters. Despite being simple and inexpensive components, 4.7k Ω resistors are critical for the proper functioning and reliability of the entire health monitoring system.

4.1.9 CONNECTING WIRES

Connecting wires are essential components used to establish electrical connections between various modules in the system, such as the ESP32 microcontroller, MAX30102 sensor, DS18B20 temperature sensor, Micro SD card module, power supply, and other peripherals. These wires carry both power and data signals, ensuring proper communication and operation of the entire circuit. In the prototyping stage, jumper wires are commonly used for flexibility and easy modification, while in the final design, PCB traces or soldered connections provide a more stable and permanent solution.



Figure 4.9 Connecting wires

The quality and arrangement of connecting wires play a significant role in system performance. Proper wiring helps reduce signal loss, noise interference, and connection errors, especially in communication protocols like I²C and one-wire. Secure and well-organized connections ensure accurate sensor readings and prevent system failures caused by loose or improper contacts. Additionally, using appropriate wire lengths and proper insulation improves reliability and safety, making connecting wires a crucial element in the successful implementation of the health monitoring system.

4.1.10 10 μ F CAPACITORS

The 10 μ F capacitors are used in the circuit for power supply filtering and voltage stabilization. They act as energy storage components that help maintain a steady voltage level by supplying charge when there is a sudden drop in voltage and absorbing excess charge during voltage spikes. These capacitors are typically placed near the ESP32 microcontroller and sensor modules such as MAX30102 to ensure a stable power supply during operation, especially when the system experiences variations in current demand.



Figure 4.10 10 μ F capacitors

These capacitors play an important role in reducing voltage ripple and electrical noise in the circuit, which is essential for accurate sensor readings and reliable system performance. In wireless communication systems like ESP32-based designs, sudden current changes can occur during data transmission, and the 10 μ F capacitors help handle these fluctuations effectively. By improving power stability, they prevent unexpected resets, signal distortion, and measurement errors. Their simple design, low cost, and high effectiveness make them a crucial component in ensuring smooth and continuous operation of the health monitoring system.

4.1.11 0.1 μ F CERAMIC CAPACITORS

The 0.1 μ F ceramic capacitors are used in the circuit for high-frequency noise filtering and decoupling purposes. They are typically placed close to the power pins of the ESP32, MAX30102 sensor, and other electronic components to filter out high-frequency electrical noise and ensure a clean power supply. These capacitors respond quickly to rapid voltage changes, making them effective in stabilizing signals at high frequencies.



Figure 4.11 0.1 μ F Ceramic Capacitors

They play a crucial role in improving the accuracy and reliability of the system by preventing noise interference that could affect sensor readings and communication signals. In digital circuits, especially with microcontrollers and communication protocols like I²C, noise can cause errors or unstable behavior. The 0.1 μ F capacitors help maintain stable voltage levels and smooth operation by eliminating unwanted disturbances. Their small size, low cost, and high efficiency make them essential components for proper circuit performance in embedded systems.

4.2 SOFTWARE

4.2.1 Arduino IDE

The Arduino Integrated Development Environment (Arduino IDE) is an open-source software platform used for writing, compiling, and uploading code to microcontrollers such as the ESP32 used in this project. It provides a simple and user-friendly interface, making it suitable for developing embedded and IoT applications. In this system, the Arduino IDE is used to program the ESP32 to collect data from sensors like the MAX30102 and DS18B20, process the data, and control modules such as the display, SD card, and WiFi communication. The IDE supports programming in C/C++ and offers a wide range of built-in and external libraries that simplify interfacing with sensors and communication protocols like I²C and one-wire. It also includes features such as code verification, error detection, and a serial monitor for real-time debugging and testing. By installing the ESP32 board package, the IDE allows seamless programming and deployment. Its cross-platform compatibility and ease of use make it an essential tool for implementing the Compact Health Parameter Recorder.

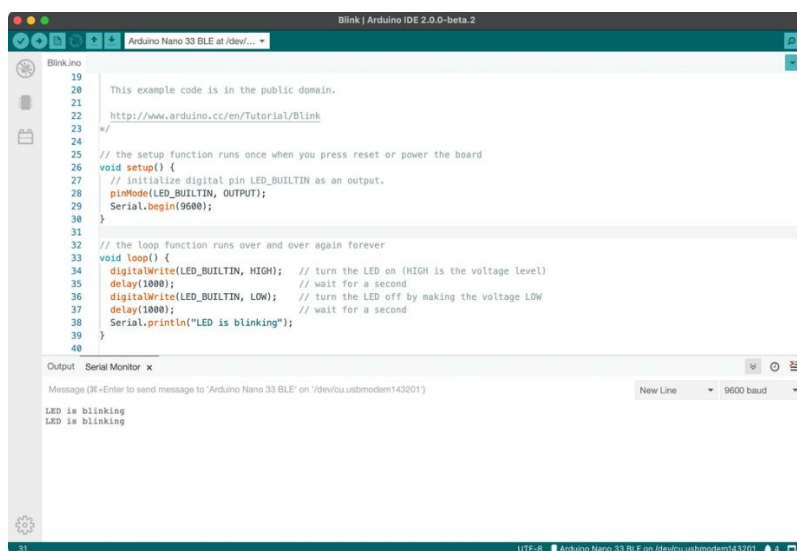


Figure 4.12 Arduino IDE

4.2.2 API PERPLEXIVITY

The Perplexity API is an AI-powered interface used to analyze data and generate intelligent insights using natural language processing techniques. In this project, it can be used to process collected health parameters such as heart rate, SpO₂, body temperature, and fatigue score to generate meaningful summaries and health reports. It helps convert raw sensor data into easy-to-understand information by identifying abnormal trends, detecting potential health risks, and providing useful recommendations. By integrating the API, the system can offer real-time analysis, automated decision-making, and alert notifications without requiring heavy processing on the ESP32. This reduces computational load on the device while enhancing system intelligence. Additionally, it improves user experience by presenting data in a clear and informative manner, making it easier for users and healthcare professionals to understand the patient's condition. Overall, the Perplexity API adds an advanced analytical layer to the system, making it more efficient, smart, and suitable for modern IoT-based healthcare applications.



Figure 4.13 API PERPLEXIVITY

4.2.3 PYTHON FLASK

Python Flask is a lightweight web framework used to build web applications and APIs for handling data communication and visualization. In this project, Flask is used to create a backend server that receives health data from the ESP32 through WiFi, processes it, and displays it on a web dashboard. It allows easy handling of HTTP requests, enabling real-time data transfer between the hardware system and the user interface. The collected data such as heart rate, SpO₂, temperature, and fatigue score can be stored, analyzed, and presented in a structured format using Flask. Flask also supports integration with databases and APIs, making it suitable for developing scalable IoT applications. It enables features like real-time monitoring, data visualization using graphs, and alert generation for abnormal conditions. The simplicity, flexibility, and minimal resource requirements of Flask make it ideal for rapid development of web-based health monitoring systems. Overall, Flask acts as a bridge between the hardware and the user, providing an interactive platform for accessing and analyzing health data.



Figure 4.14 Python Flask

CHAPTER 5

RESULTS & DISCUSSION

The developed Compact Health Parameter Recorder was successfully implemented and tested for measuring vital parameters such as heart rate, SpO₂, and body temperature. The MAX30102 sensor provided accurate and consistent heart rate and oxygen saturation readings when proper finger placement was maintained. The DS18B20 sensor delivered stable and reliable temperature values with minimal deviation. The ESP32 effectively processed the sensor data in real time, displaying the results on the user interface and storing them in the SD card for long-term analysis. The system also demonstrated successful wireless transmission of data through WiFi for remote monitoring.

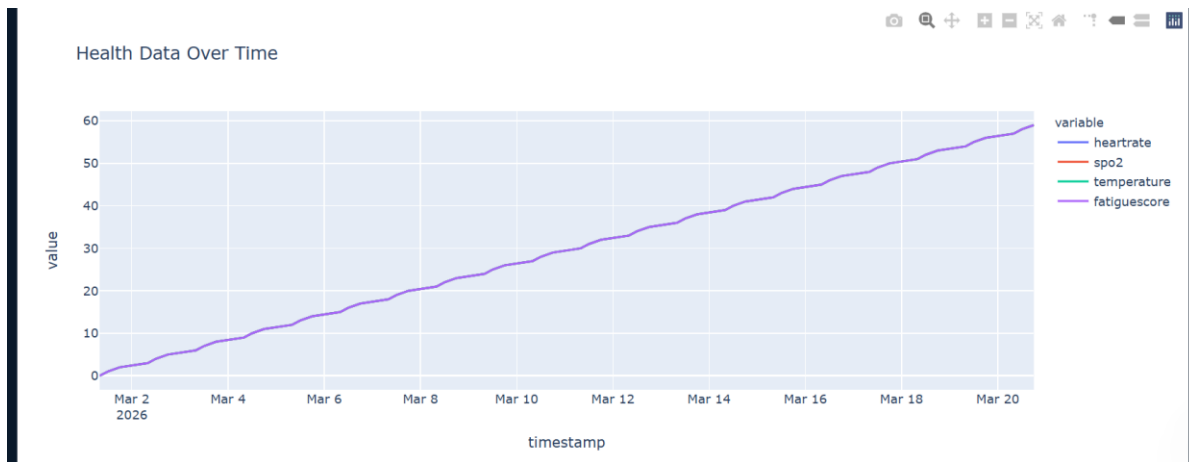


Figure 5.1 Results of Proposed Method

The overall performance of the system was found to be efficient, compact, and suitable for continuous monitoring applications. Minor variations in readings were observed due to motion artifacts and improper sensor contact, which can be minimized with proper usage. The power consumption was optimized using low-power features of the ESP32, allowing extended operation with the Li-Po battery. The system proved to be

reliable for tracking health trends over time and can be effectively used for fatigue detection and preventive healthcare applications.

Graph Analysis

- The x-axis represents time (timestamp/dates) over the monitoring period.
- The y-axis represents the values of health parameters such as heart rate, SpO₂, temperature, and fatigue score.
- The graph shows a gradual increasing trend in the values over time.
- The fatigue score increases continuously, indicating rising fatigue levels.
- The curve is smooth without sudden spikes, showing stable and reliable sensor performance.
- The data indicates consistent monitoring and accurate data collection.
- The trend helps in identifying gradual health changes and supports early fatigue detection.

The graph illustrates the variation of health parameters over time, where the x-axis represents the timestamp (dates) and the y-axis represents the measured values such as heart rate, SpO₂, temperature, and fatigue score. It shows a gradual and consistent increase in the values, with the fatigue score rising steadily, indicating increasing fatigue levels over the monitored period. The smooth and continuous nature of the curve indicates stable sensor performance, accurate data collection, and minimal noise interference. The absence of sudden fluctuations suggests reliable system operation. Overall, the graph highlights the effectiveness of continuous monitoring in identifying gradual physiological changes and supports early detection of fatigue and preventive healthcare management.

Data					
timestamp	heartrate	spo2	temperature	fatiguescore	danger
2026-03-01 08:00:00	72	97	36.7	0.0	False
2026-03-01 12:00:00	75	96	36.8	0.0	False
2026-03-01 18:00:00	78	95	37.0	0.0	False
2026-03-02 08:00:00	80	94	37.1	0.6	False
2026-03-02 12:00:00	130	85	38.5	7.8	True
2026-03-02 18:00:00	85	92	37.2	0.7	False
2026-03-03 08:00:00	88	91	37.3	1.2	False
2026-03-03 12:00:00	95	89	38.1	5.4	True
2026-03-03 18:00:00	100	93	37.5	0.0	False
2026-03-04 08:00:00	110	92	37.6	1.7	False
2026-03-04 12:00:00	115	95	36.9	0.8	False
2026-03-04 18:00:00	120	94	37.2	1.1	False
2026-03-05 08:00:00	125	88	38.7	9.0	True
2026-03-05 12:00:00	118	91	37.8	2.6	False
2026-03-05 18:00:00	112	90	37.5	1.5	False
2026-03-06 08:00:00	105	93	37.0	0.3	False
2026-03-06 12:00:00	100	92	36.8	0.0	False
2026-03-06 18:00:00	102	90	37.1	0.4	False
2026-03-07 08:00:00	95	95	36.7	0.0	False

Figure 5.2 Stored data of the patient

The table presents the recorded health data over a period of time, including parameters such as timestamp, heart rate, SpO₂, body temperature, fatigue score, and danger indication. The timestamp column shows the date and time of each measurement, indicating that data is collected at regular intervals throughout the day. The heart rate values range from normal resting levels to higher values, while SpO₂ levels mostly remain within a healthy range with slight variations. Temperature readings are generally stable around normal body temperature but show occasional increases, which may indicate physical stress or fatigue. The fatigue score varies based on the changes in these parameters, and the danger column flags abnormal conditions when certain thresholds are exceeded.

From the table, it is observed that higher fatigue scores are associated with increased heart rate, reduced SpO₂ levels, and elevated temperature values. For example, entries with higher heart rate and temperature often correspond to higher fatigue scores and are marked as “True” in the danger column. This indicates that the system is effectively identifying potential health risks based on combined parameter analysis. The consistent data logging demonstrates the system’s capability for continuous monitoring, while the variation in values highlights its usefulness in detecting changes in physiological conditions. Overall, the table shows that the system can accurately record, analyze, and flag health data, supporting early fatigue detection and preventive healthcare.

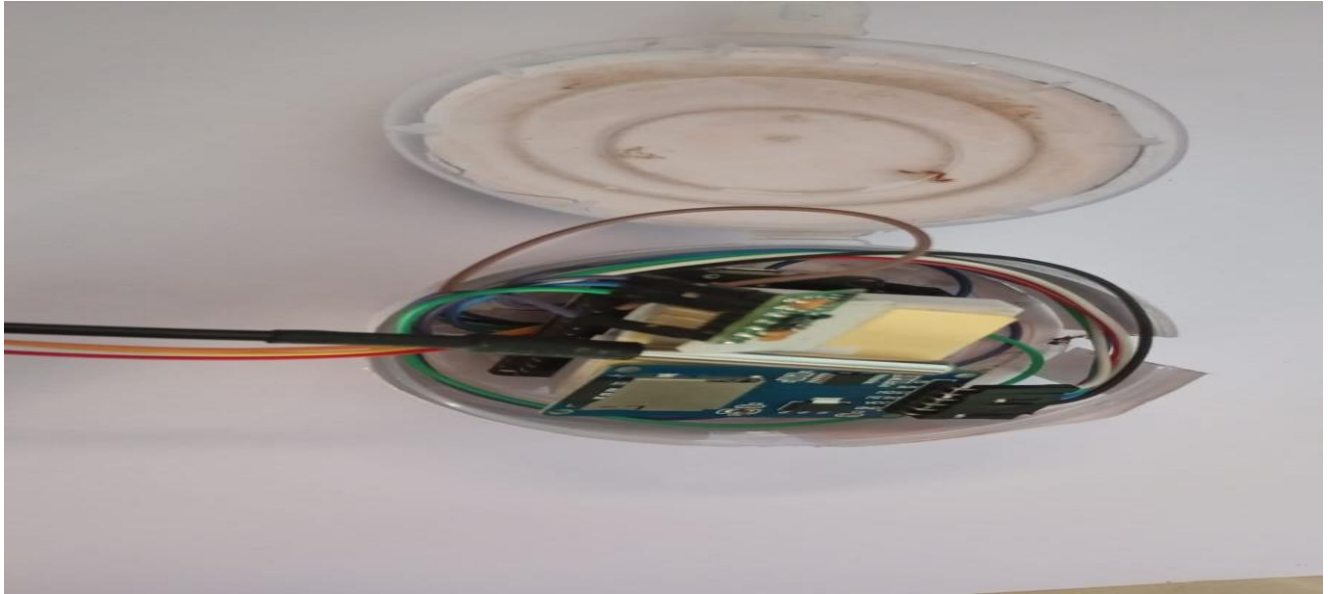


Figure 5.3 Working Model

The Compact Health Parameter Recorder is a wearable system designed to continuously monitor vital health parameters such as heart rate, blood oxygen saturation (SpO_2), and body temperature. It uses an ESP32 microcontroller along with sensors like MAX30102 and DS18B20 to collect and process real-time physiological data. The system stores the data for long-term analysis and also transmits it wirelessly for remote monitoring. It helps in detecting fatigue and abnormal health conditions at an early stage. The compact design, low power consumption, and cost-effectiveness make it suitable for preventive healthcare, remote patient monitoring, and wearable health applications.

CHAPTER 6

CONCLUSION

The Compact Health Parameter Recorder developed in this project successfully demonstrates an efficient, reliable, and cost-effective system for continuous monitoring of vital health parameters such as heart rate, blood oxygen saturation (SpO₂), and body temperature. By integrating sensors like MAX30102 and DS18B20 with the ESP32 microcontroller, the system is capable of real-time data acquisition, processing, and analysis. The inclusion of SD card storage enables long-term data logging, while WiFi connectivity allows remote monitoring and accessibility through IoT platforms. The system effectively tracks variations in physiological parameters and provides meaningful insights into fatigue levels, making it highly suitable for preventive healthcare, wearable applications, and remote patient monitoring. Its compact size, low power consumption, and ease of implementation further enhance its practicality for real-world deployment.

The experimental results show that the system provides stable, accurate, and consistent measurements under normal operating conditions. The data visualization through graphs, tables, and dashboard analysis enhances understanding by clearly showing trends and highlighting abnormal conditions. The fatigue detection mechanism, based on combined parameter analysis, proves useful in identifying early signs of stress and health risks. Although minor limitations such as motion artifacts and sensor placement sensitivity exist, they can be minimized with proper usage and improved sensor placement techniques. With future enhancements such as AI-based analysis, additional sensors, and improved wearable design, the system can be made more advanced and user-friendly. Overall, the project highlights the importance of compact, low-power, and IoT-enabled health monitoring systems, contributing to early diagnosis, continuous health tracking, and improved quality of life.

6.1 FUTURE ENHANCEMENTS

- **Integration of Advanced Sensors:** Additional sensors such as ECG and respiratory sensors can be added to provide more detailed health analysis and improve fatigue detection accuracy. These sensors can help monitor cardiac activity and breathing patterns, giving a more complete understanding of the user's health condition. This will enhance the system's capability in detecting critical health issues at an early stage.

- **AI-Based Health Analysis:** Machine learning algorithms can be implemented to analyze data patterns and provide accurate fatigue prediction and personalized health recommendations. AI can learn from historical data and improve prediction accuracy over time. This will make the system smarter and capable of automatic decision-making without constant human intervention.

- **Cloud and Mobile Application Integration:** Developing a mobile app and cloud platform can enable real-time monitoring, data visualization, and alert notifications for users and doctors. It allows easy access to health data from anywhere and supports remote healthcare services. This improves convenience, data sharing, and continuous monitoring.

- **Power Optimization:** Advanced power management techniques and energy-efficient components can be used to extend battery life for long-term continuous monitoring. Methods like sleep modes, duty cycling, and low-power sensors can reduce energy consumption. This makes the device more suitable for wearable and portable applications.

- **Improved Wearable Design:** Enhancing the device design for better comfort, compactness, and reduced motion artifacts will improve usability and accuracy in real-world applications. A more ergonomic design ensures better user experience during long-term usage. It also helps in maintaining proper sensor contact for accurate readings.

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